

Exercise 10.1

Determine whether the following systems are observable:

$$\begin{cases} \dot{x} = \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix} x \\ y = \begin{bmatrix} 1 & 0 \end{bmatrix} x \end{cases} \quad \begin{cases} \dot{x} = \begin{bmatrix} -1 & 0 \\ 1 & -1 \end{bmatrix} x \\ y = \begin{bmatrix} 1 & 0 \end{bmatrix} x \end{cases}$$

$$\begin{cases} \dot{x} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} x \\ y = \begin{bmatrix} 1 & 1 \end{bmatrix} x \end{cases} \quad \begin{cases} \dot{x} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} x \\ y = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} x. \end{cases}$$

Exercise 10.2

Consider the following system from Exercise 9.3

$$\begin{cases} \dot{x} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} u \\ y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} x, \end{cases}$$

and apply the control $u = Fx$ (designed in Exercise 9.3) yielding closed loop poles -1 , -2 , and -3 .

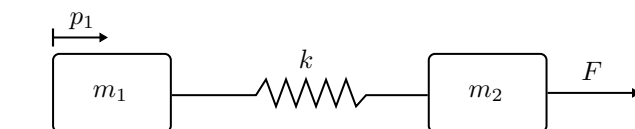
1. Decide where the observer eigenvalues should be placed, and argue why this is an appropriate choice.
2. Design an observer that complies with your particular choice of eigenvalues.
3. Repeat the observer design using `place` with the following output equation

$$y = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} x.$$

4. Draw a diagram of the system, observer, and feedback.
5. Calculate the eigenvalues of the closed loop system, and argue why they are placed here.
6. Simulate the entire system, and compare your state estimate with the correct state.

Exercise 10.3

Consider the model from Exercise 2.1 of two masses connected via a lossless spring, moving at a surface with no friction, shown in the figure below.



1. Determine which measurements or combinations of measurements that render the system observable.